PARTS LIST

Item	Qty	REFERENCE DESIGNATOR	PART NUMBER	Description	MANUFACTURER
1	1		S1101EFF0302	BARE BOARD	ZIN TECHNOLOGIES
2	20	C1, C2, C3, C4, C5, C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C17, C18, C19, C20	C0805C104K1RACTU	CAPACITOR 0.1uF 100V CERAMIC 10% 0805 SMD	KEMET
3	3	D1, D2, D3	SBR3U60P1	DIODE, RECTIFIER, 3A, 60V, POWERDI123	DIODES, INC
4	1	J9	SDD78M3S00G	CONN, SUB-D, 78 PIN, MALE, VERT	POSITRONICS
5	1	J10	SND25M3S00G	CONN, SUB-D, 25 PIN, MALE, VERT	POSITRONIC
6	2	J69, J72	SDD44M3S00G	CONNECTOR, SUB-D, 44 PIN, 3 ROW, MALE, VERTICAL WITH SWAGED SPACERS	POSITRONIC
7	1	J83	SDD26M3S00G	CONNECTOR, SUB-D, 26 PIN, 3 ROW, MALE, VERTICAL WITH SWAGED SPACERS	POSITRONIC
8	1	J100	SND9M3S00G	CONNECTOR, SUB-D, 9-PIN MALE VERTICAL WITH SPACERS 4-40	POSITRONIC
9	1	P1	ESW-120-12-L-D	CONNECTOR, ELEVATED SOCKET, .1 PITCH	Samtec
10	20	R1, R5, R7, R8, R9, R13, R14, R15, R16, R21, R25, R26, R32, R34, R35, R37, R40, R41, R44, R51	ERJ-6ENF22R1V	RESISTOR 22.1 OHM 1/8W 1% 0805 SMD	PANASONIC - ECG
11	3	R2, R46, R47	ERJ-6GEY0R00V	RESISTOR 0.0 OHM 1/8W 5% 0805 SMD	PANASONIC - ECG
12	25	R3, R4, R6, R10, R11, R12, R17, R18, R19, R20, R23, R24, R28, R29, R30, R33, R36, R38, R39, R42, R43, R45, R49, R50, R52	ERJ-6ENF1002V	RESISTOR 10.0K 1/8W 1% 0805 SMD	PANASONIC - ECG
13	3	R22, R27, R31	ERJ-6ENF1211V	RESISTOR, 1.21K, 1/8W, 1%, 0805 SMD	PANASONIC - ECG
14	A/R		TZE-211	LABEL, 1/4 BLK LETTERS WHT BACKROUND	BROTHER
15	A/R		3140	RTV FOR CONFORMAL COATING	DOW CORNING
16	A/R		PR-1204	PRIMER FOR CONFORMAL COATING	DOW CORNING
17	A/R		SN63PB37/RMA	SOLDER	
18	4		JAX108-M09-F09-440	JACK SCREW, .112-40 UNC-2AX.625 LG THREADED, 303 STAINLESS STEEL, PASSIVATED	UNICOROP INC
19	36		NAS620C4	#4 FLAT WASHER REDUCED OUTER DIAMETER	QPL
20	12		MS21043-04	#4-40 UNJC-3B REDUCED HEX SELF LOCKING REDUCED HEIGHT NUT	QPL
21	8		JAX110-M09-F09-440	JACK SCREW, .112-40 UNC-2AX.750 LG THREADED, 303 STAINLESS STEEL, PASSIVATED	UNICOROP INC

8. CONFORMAL COATING, STAKING AND INSPECTION SHALL BE IN ACCORDANCE WITH NASA-STD-8739.1 AND THE FOLLOWING NOTES:
   - A. PERFORM CLEANLINESS TEST.
   - B. BAKE PWA AT 93°C ±5.5°C (200°F ±10°F) FOR 4 HOURS PRIOR TO CONFORMAL COATING.
   - C. MASK OFF ALL DESIGNATED AREAS USING KAPTON TAPE. ALL MASKING OPERATIONS SHALL BE PERFORMED UNDER AN AIR IONIZER.
   - D. MEASURE CONFORMAL COATING THICKNESS AT DESIGNATED AREAS.
   - E. CONFORMAL COAT THE COMPLETE ASSEMBLY USING ITEM 15 AND ITEM 16.
9. ENGINEER OR TECHNICIAN SHALL FUNCTIONALLY TEST THE COMPLETED ASSEMBLY AFTER CONFORMAL COATING PER TEST PLAN ACME-TST-031.
10. PLACE THE COMPLETED ASSEMBLY IN CLEAN, STATIC DISSIPATIVE PACKAGING. IDENTIFY PART BY TAGGING WITH PART NUMBER AND SERIAL/LOT NUMBER AS SPECIFIED IN WORK ORDER PER SAE AS478, METHOD 35. STORE IN A PROJECT APPROVED STORAGE AREA.
11. SNUG TIGHT:
    FOR FASTENERS #6 AND BELOW: AFTER REMOVING ALL GAPS BETWEEN CLAMPED COMPONENTS, FASTENERS SHALL HAVE AN ADDITIONAL 1/8 TURN.
12. REFERENCE SCHEMATIC: S1101EFS0302
13. EXPEDITION CONTROL FILES:
    S1101EFA0302_REV_C.ZIP

NOTES: UNLESS OTHERWISE SPECIFIED

1. CERTIFICATE OF CONFORMANCE REQUIRED FOR ALL COMMERCIAL PARTS.
2. ESD HANDLING AND CONTROL PER ESTABLISHED PROCEDURES.
3. HANDLE ALL PARTS AND ASSEMBLIES WITH CLEAN, LINT-FREE, ANTI-STATIC GLOVES OR FINGER COTS.
4. DEMOISTURIZE (BAKE OUT) PWB (ITEM 1) FOR A MINIMUM OF 2 HOURS AT 93°C ±5.5°C (200°F ±10°F.) PERFORM SOLDERING IMMEDIATELY AFTER BAKING, OR STORE IN A CLEAN, DRY, SEALED STATIC DISSIPATIVE BAG OR CONTAINER.
5. MARK THE ASSEMBLY WITH REV AND S/N USING ITEM 14.
6. ASSEMBLY, SOLDERING AND INSPECTION SHALL BE IN ACCORDANCE WITH NASA-STD-8739.2, NASA-STD-8739.3, AND THE FOLLOWING NOTES:
   - A. INSTALL AND SOLDER ALL SURFACE MOUNT (SMT) PART BEFORE THROUGH HOLE PARTS USING SOLDER (ITEM 17.)
   - B. FOR THROUGH HOLE PARTS, LEAD TERMINATIONS SHALL BE STRAIGHT THROUGH (NOT LAPPED OR CLINCHED.)
   - C. DISCARD WASHERS AND NUTS FORM ITEMS 18 AND 21.
7. ENGINEER OR TECHNICIAN SHALL FUNCTIONALLY TEST THE COMPLETED ASSEMBLY PRIOR TO CONFORMAL COATING PER TEST PLAN ACME-TST-031.

REVISION HISTORY

REV	DESCRIPTION	APPROVED
C	REVISE PER ACME-ECO-373	MH 3/18/15 TG 3/19/15 DS 3/19/15
B	REVISE PER ACME-ECO-315 AND ACME-ECO-328	TAG 9/16/14 DAS 9/16/14 RJP 9/16/14
A	REVISE PER ACME-ECO-295	TAG 4/14/14 DAS 4/10/14 RJP 4/10/14

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S1101MFA15200314

NEXT ASSEMBLY

UNLESS OTHERWISE SPECIFIED:
1. DIMENSIONS ARE IN INCHES
2. BREAK EDGES .010-.020
3. INSIDE CORNERS R.02 MAX
4. LINEAR TOLERANCES:
   .X ± 0.03
   .XX ± 0.01
   .XXX ± 0.005
5. ANGULAR TOLERANCES: ± 0.5°

THIRD ANGLE PROJECTION

THIS DRAWING IS UNDER CONFIGURATION CONTROL: DO NOT REVISE WITHOUT AN APPROVED ECO.

APPROVED: E. McCLOSKEY 3/19/15

PRIOR TO USE, CHECK WITH CONFIGURATION MANAGEMENT FOR ANY OUTSTANDING ECOS.

DESIGNER	R. PFEIFFER 7/30/13	QUALITY	D. CHOVAN 7/31/13
REVIEWER	D. STRANG 7/31/13	MATERIALS	M. WILKINSON 8/5/13
MECHANICAL	A. DRAKE 7/31/13	PROJ LEAD	B. BOROWSKI 8/19/13
ELECTRICAL	T. GOBELI 7/31/13	-	-
STRUCTURAL	W. BORRELLI 7/301/13	-	-
SAFETY	C. WERNER 7/31/13	-	-

DIMENSIONING AND TOLERANCING PER ASME Y14.5M-1994

CONTRACT NUMBER: NNC09BA02B
DELIVERY CODE: SPACE110

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NATIONAL AERONAUTICS AND SPACE ADMINISTRATION
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LEWIS FIELD
CLEVELAND, OHIO

TITLE
ACME
UPSTREAM RELAY CONNECTOR BOARD
ASSEMBLY DRAWING

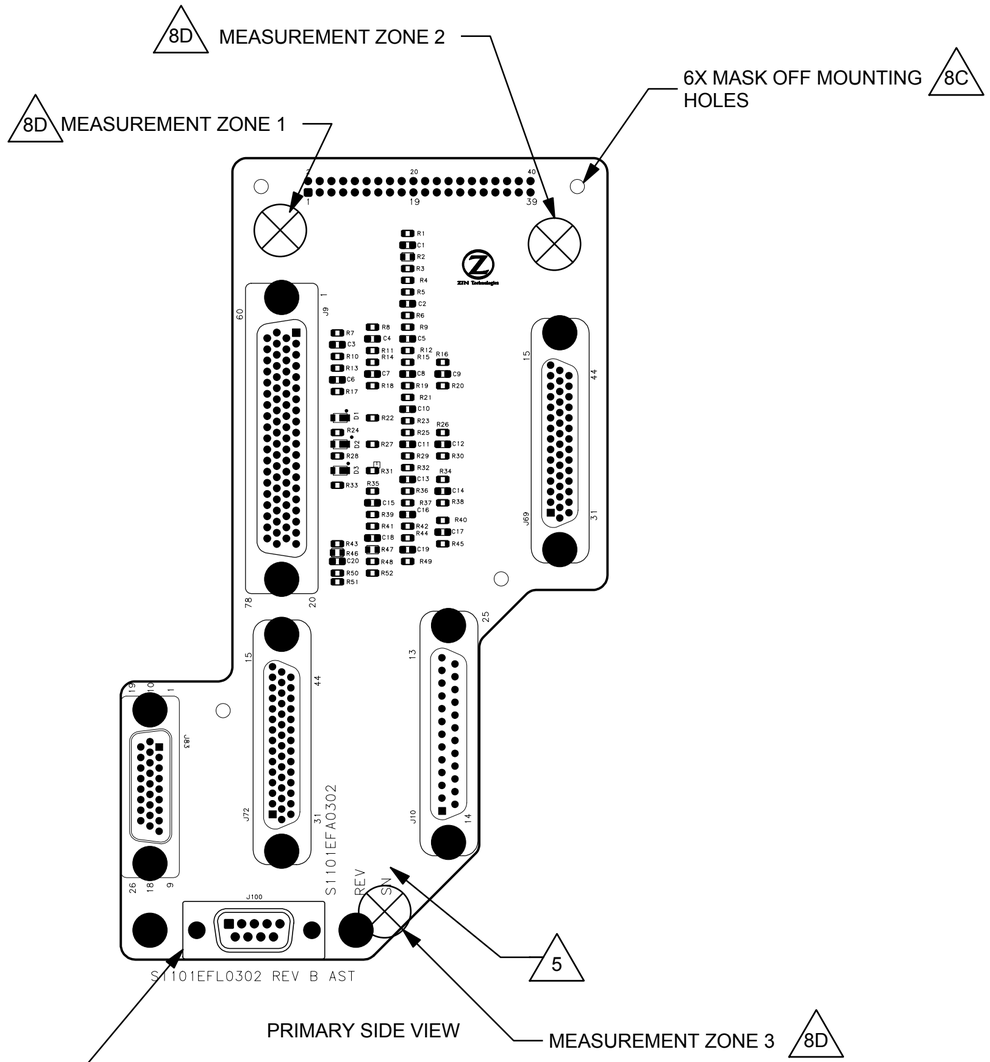
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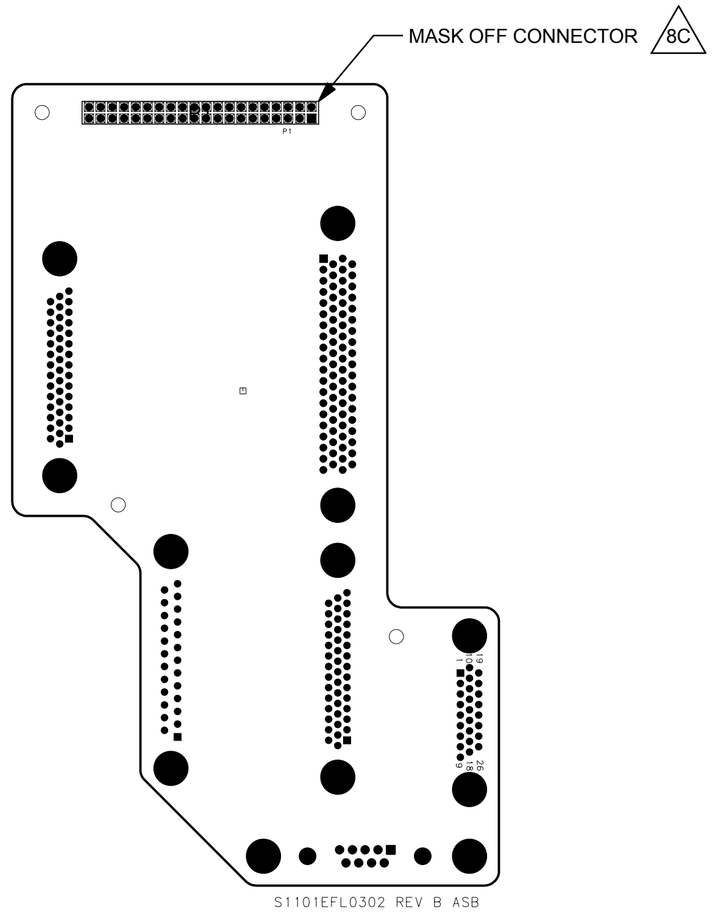
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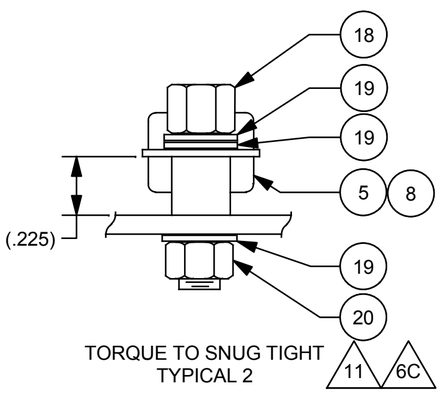
RELEASE LEVEL: .
MODEL NAME: S1101EFA0302_REV_C.bpd

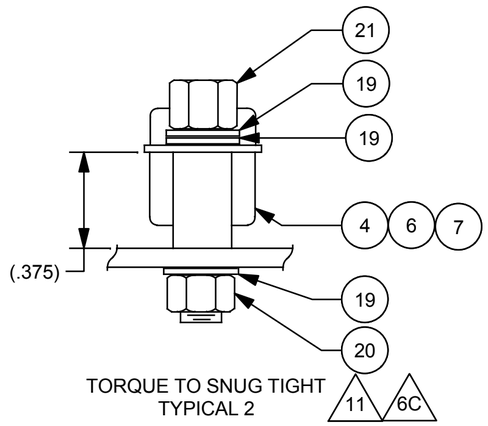
BLUEPRINT
VERSION 4.0

REV C
SH 1
DWG NO S1101EFA0302









SIZE	CAGE CODE	DWG NO	REV
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SCALE: NONE	INITIAL REL DATE 8/19/13	SHEET 2 OF 2	

BLUEPRINT
VERSION 4.0

REV C
SH 2
DWG NO S1101EFA0302